

A One-Point Primitive Space Need Not Give BCP

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving the commutative unital C^* -algebra case, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

The packet `2311.14257_primitive_pi_base_obstruction` proves that if a C^* -algebra A has BCP, then $\text{Prim}(A)$ has a countable π -base. The following observation shows that this necessary condition is not sufficient, even in the smallest possible primitive topology.

Theorem 1. *Let $\mathcal{M}$ be any II_1 factor, for example the hyperfinite II_1 factor $\mathcal{R}$. Then*

$$\text{Prim}(\mathcal{M}) = \{0\},$$

so $\text{Prim}(\mathcal{M})$ has a countable π -base, but $\mathcal{M}$ fails BCP.

Corollary 1. *The implication*

$$\text{Prim}(A) \text{ has a countable } \pi\text{-base} \implies A \text{ has BCP}$$

is false for C^ -algebras. It is false even among C^* -algebras with one-point primitive ideal space.*

Proof Intuition

The primitive ideal space only records closed two-sided ideals visible through irreducible representations. A II_1 factor has no nontrivial closed two-sided C^* -ideals, so its primitive ideal space is a singleton. This topological smallness does not imply BCP. Huang–Kudaybergenov–Liu prove that atomless von Neumann algebras fail BCP, and a II_1 factor is atomless. Thus the primitive topology misses a genuinely noncommutative obstruction: finite diffuse projection structure.

The Simplicity Lemma

Lemma 1. *Every II_1 factor $\mathcal{M}$ is simple as a C^* -algebra.*

Proof. Let I be a nonzero norm-closed two-sided ideal in $\mathcal{M}$. Choose $0 \neq a \in I_+$. Pick $\delta > 0$ such that the spectral projection

$$p = 1_{[\delta, \infty)}(a)$$

is nonzero. On $p\mathcal{M}p$, the element ap is invertible and $(ap)^{-1}p \in p\mathcal{M}p$ is bounded. Since I is a two-sided ideal and $a \in I$,

$$p = ((ap)^{-1}p)ap \in I.$$

Let τ be the normalized trace of $\mathcal{M}$. Since $p \neq 0$, put $t = \tau(p) > 0$. Choose n with $1/n \leq t$, and decompose the unit as

$$1 = e_1 + \cdots + e_n, \quad \tau(e_k) = 1/n.$$

By comparison of projections in a finite factor, each e_k is Murray–von Neumann subequivalent to p . Hence there are projections $q_k \leq p$ and partial isometries $v_k \in \mathcal{M}$ such that

$$v_k^*v_k = e_k, \quad v_kv_k^* = q_k.$$

Because $p \in I$, also $q_k = pq_kp \in I$, and therefore

$$e_k = v_k^*q_kv_k \in I.$$

Summing over k gives $1 \in I$. Thus $I = \mathcal{M}$, and $\mathcal{M}$ is simple. □

Proof of the Obstruction

Proof of Theorem 1. By Lemma 1, $\mathcal{M}$ is simple as a C^* -algebra. A nonzero C^* -algebra has an irreducible representation; since $\mathcal{M}$ is simple, the kernel of any irreducible representation is the only proper closed two-sided ideal, namely 0. Hence 0 is primitive and

$$\text{Prim}(\mathcal{M}) = \{0\}.$$

This one-point space has a countable π -base.

On the other hand, $\mathcal{M}$ is a nontrivial atomless von Neumann algebra: as a II_1 factor it has no minimal projections. The source paper proves that every nontrivial atomless von Neumann algebra fails BCP [1, Theorem labelled atomlessnotbc; equivalently by the main von Neumann classification, Theorem labelled main1]. Therefore $\mathcal{M}$ fails BCP. □

Proof of Corollary 1. Apply Theorem 1. □

Relation to the Primitive Pi-Base Packet

The earlier primitive-ideal theorem remains a broad necessary condition:

$$A \text{ has BCP} \implies \text{Prim}(A) \text{ has a countable } \pi\text{-base.}$$

The present packet shows that the converse cannot be the desired noncommutative analogue of the commutative $C(K)$ theorem. Any full classification must add information invisible to $\text{Prim}(A)$, at least enough to distinguish diffuse finite factors from atomic type I von Neumann algebras.

Verification and Search Bounds

The run indexes were searched for arXiv:2311.14257 together with II_1 factor, factor, primitive ideal, one-point primitive space, countable π -base, BCP, and UBCP. Existing packets contained the primitive pi-base necessary condition and separable/countably generated positive result, but no packet recording that the necessary condition is strict.

The proof uses the source paper's von Neumann algebra theorem for BCP failure of atomless von Neumann algebras. The C^* -simplicity of II_1 factors is included for self-containment, using standard spectral cutoffs and projection comparison in finite factors. This packet is therefore best read as a sharp obstruction to a candidate characterization, not as an independent new proof of the source paper's atomless von Neumann theorem.

Human Review Focus

Check Lemma 1, especially the passage from a nonzero positive element in an ideal to a nonzero spectral projection and the finite projection-comparison step. Then verify that the source theorem indeed applies to II_1 factors.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Masamichi Takesaki, *Theory of Operator Algebras I*, Springer, 1979.
- [3] Jacques Dixmier, *C^* -algebras*, North-Holland, 1977.